

Dawei Wang

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EDUCATION

Honours Bachelor of Science - University of Toronto

Sep. 2022 - Jun. 2026

- Programs: Immunology Major, Health and Disease Major

SKILLS

Programming: MATLAB, Python, R, FreeSurfer, BIDS, Bash

Laboratory: Mammalian cell culture, transfection, mouse handling, immunohistochemistry, Western blotting, bacterial cell culture, transformation, plasmid isolation, RT-qPCR, cloning (Golden-Gate, Gibson), fluorescence microscopy

RESEARCH EXPERIENCES

Master's of Science Candidate - Centre for Addiction and Mental Health (CAMH)

Sep. 2026 - Present

- Project: Characterization and Prediction of Individual Functional Outcome Trajectories in Schizophrenia Spectrum Disorders (PREDICTS Study)
- Supervisor: Dr. Nicholas Neufeld
- Description:
 - Provided questionnaires to schizophrenia spectrum disorder patients to evaluate their psychosocial well-being, behavioural/task performance, and quality of life
 - Analyzed patient fMRI data using FreeSurfer to search for possible patterns and biomarkers in connectivity networks
 - Observed grey matter microstructure data in schizophrenia patients; correlated changes in brain structure with metabolic changes

Thesis Student - Hospital for Sick Children (SickKids)

Sep. 2025 - Apr. 2026

- Project: Functional Network Disruption as Biomarkers for Adolescent Concussion
- Supervisor: Dr. Benjamin Dunkley
- Description:
 - Used magnetoencephalography (MEG) and MATLAB, Python, & graph theory-based modelling to analyze resting-state intrinsic neural activity in adolescent concussion patients to identify biomarkers for concussion and improve non-invasive diagnosis methods
- Outcomes:
 - Alpha frequency connectivity measures (efficiency, strength, clustering coefficient) was found to be decreased in concussion patients compared to healthy controls

- No significant changes occurred in beta frequency connectivity measures

Research Assistant - Hospital for Sick Children (SickKids)

May 2025 - Aug. 2025

- Project: Pilot study investigating the role of glutamate receptors in synaptic plasticity and memory
- Supervisor: Dr. Zhengping Jia
- Description:
 - Investigating the role AMPA receptors and their mutations regulate Alzheimer's development using cell cultures
 - Quantified amyloid-beta plaque amounts in transgenic mice using immunohistochemical staining
- Techniques used: Mouse handling and surgery (injection, cardiovascular perfusion), immunohistochemical staining and imaging brain slices, Western blot, HEK293 cell culturing & transfection, fluorescent cell imaging
- Outcomes: No significant difference between AMPAR and APP protein surface expression

POSTERS AND PRESENTATIONS

Tri-Campus Undergraduate Research Conference - University of Toronto

Feb. 18, 2026

- Conference Speaker
- Title: "Alpha-Frequency Network Disruption as a Biomarker of Adolescent Concussion"

Child and Youth Mental Health Research Day - Hospital for Sick Children (SickKids)

Feb. 17, 2026

- Conference Speaker
- Title: "Alpha-Frequency Network Disruption as a Biomarker of Adolescent Concussion"

VOLUNTEER & EXTRACURRICULAR ACTIVITIES

iGEM Research Team Member - University of Toronto

Jun. 2024 - Nov. 2024

- Project Information: <https://2024.igem.wiki/utoronto/>
- Supervisor: Dr. Michael Garton (Department of Biomedical Engineering)
- Description:
 - Designed plasmid sequences and primers using large language model, cloned sequences onto vector plasmids
 - Transformed recombinant plasmids into *E. coli* and observed growth on antibiotic culture to validate plasmid
- Presented our project at an international competition involving institutions across 50+ countries
 - Placed among the Top 10, the only North American team on the list
 - Project and presentation received a Gold Medal and distinctions of 'Best Model', 'Best Entrepreneurship'

Debate Speaker - iGEM Toronto and Hart House Debate Club

Aug. 27, 2024

- Researched literature and spoke to experts in AI in medicine, antibiotics to support my position
- Spoke for affirmative side of debate on the ethics of rapid commercialization of AI in medicine in front of audience of ~25 consisting of students and some visiting faculty

HOSA 1st Place Competitor - HOSA – Future Health Professionals May 8, 2024

- Received 1st place nationally in the Medical Terminology event at the 9th annual HOSA Spring Leadership Conference, Canada's largest annual health science competition with 6,000+ students participating

Outreach Team Member - iGEM Toronto (Synthetic Biology Research Team) Mar. 2024 - Oct. 2024

- In charge of the team's Instagram (@igemtoronto) and email, planned and ran social media campaigns such as 'story takeovers', 'day in the life', and 'collaborative posts' to boost engagement, gaining 600 new followers
- Reached out to external organizations across Canadian universities to collaborate and further raise public awareness on our field of research

Clinical Support Volunteer - Mount Sinai Hospital Jan. 2024 - Dec. 2025

- Visited patients living with cognitive impairments weekly to check in on their well-being and provide social interaction
- Held activities with the patients; e.g. feeding meals, conversations, playing games, playing music to them, ensuring they were supported and helping maintain their cognitive function

Vice President of Administration - New College Student Council Jun. 2023 - Apr. 2024

- Elected to my university-affiliated college's student council to represent over 6,800 students and be the voice to represent their needs, concerns, and opinions
- Coordinated closely with other executives for student council elections and member transitions, ensuring rules were followed and these events ran as smoothly and efficiently as possible
- Managed and supervised team; supervised executive and member responsibilities and personal conduct, taking appropriate measures against any misconducts

Recognized Study Group Leader - University of Toronto May 2023 - Jun. 2023

- Facilitated weekly organized meetings and activities to review course material
- Prepared weekly studying material, topics of discussion, and activities

Social Chair - New College Residence Council Sep. 2022 - Apr. 2023

- Organized house-wide social events for students in my residence house/floor to increase community engagement among ~50 residents.
- Planned cultural events to better include residents of diverse backgrounds (e.g. Lunar New Year, Diwali).
- Helped plan residence-wide events with the rest of the wide Residence Council to help create a stronger community for the entire New College residence (600+ students)

AWARDS AND DISTINCTIONS

Dean's List - University of Toronto

Apr. 2023 - Apr. 2026

Awarded to students in good academic standing, with GPA above 3.5

Audrey Taylor Award - University of Toronto New College Registrar

Apr. 2024

Awarded to students in years 1-3 with good academic standing and significant co-curricular contribution.

Valued at \$400

Betty Yeoman Memorial Award - New College Office of Residence Life

Jun. 2023

Awarded to residents in good academic standing with significant contribution to the residence. Valued at \$1,400